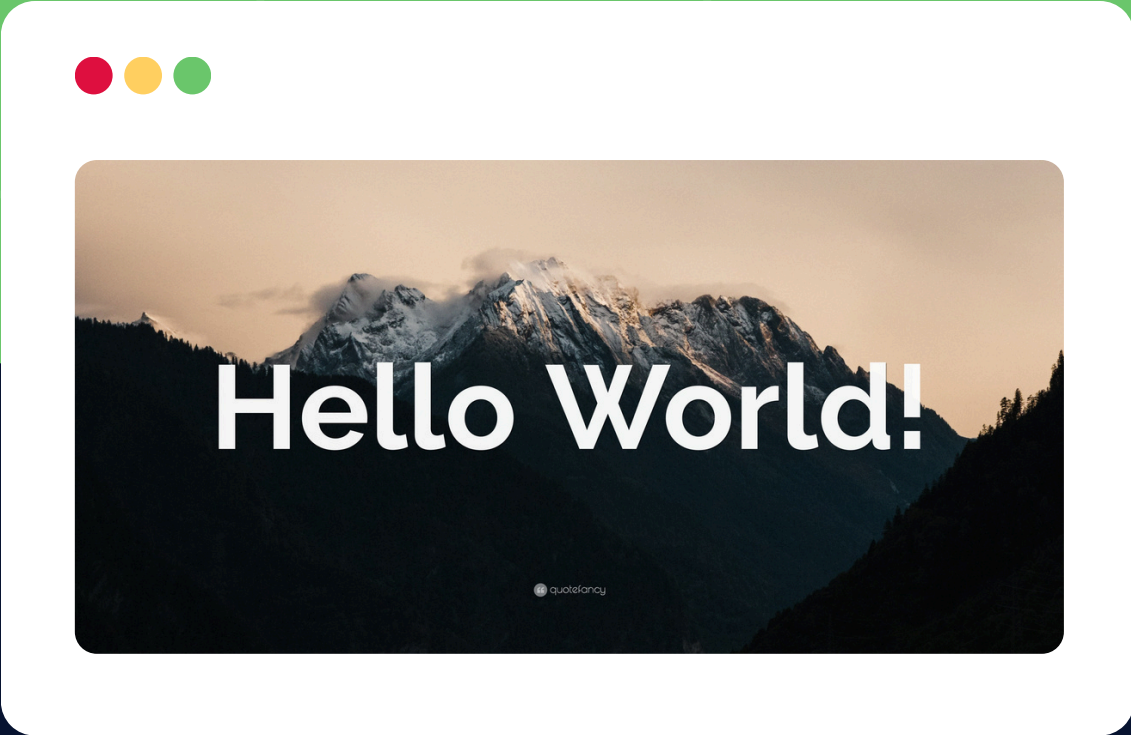
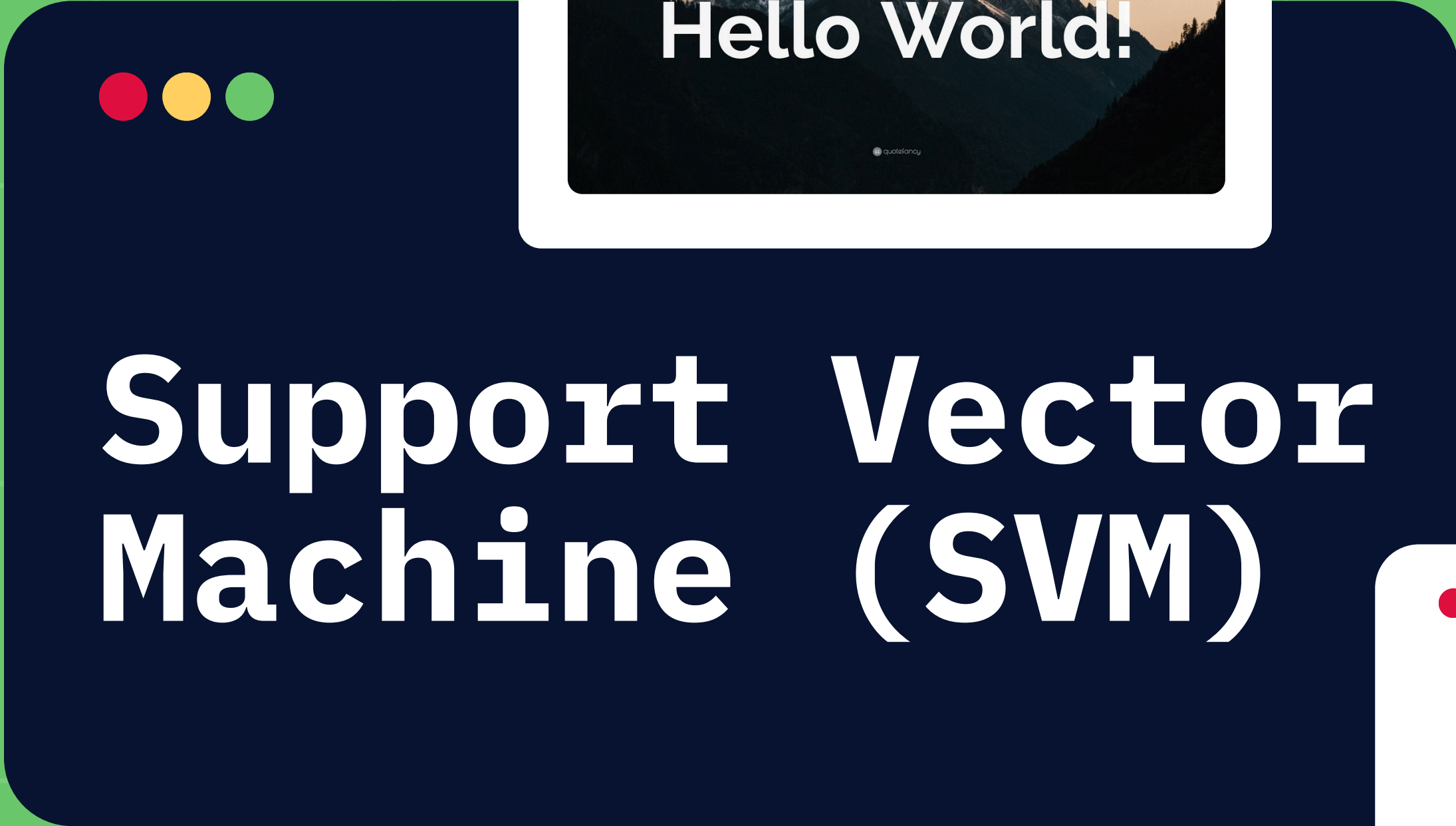
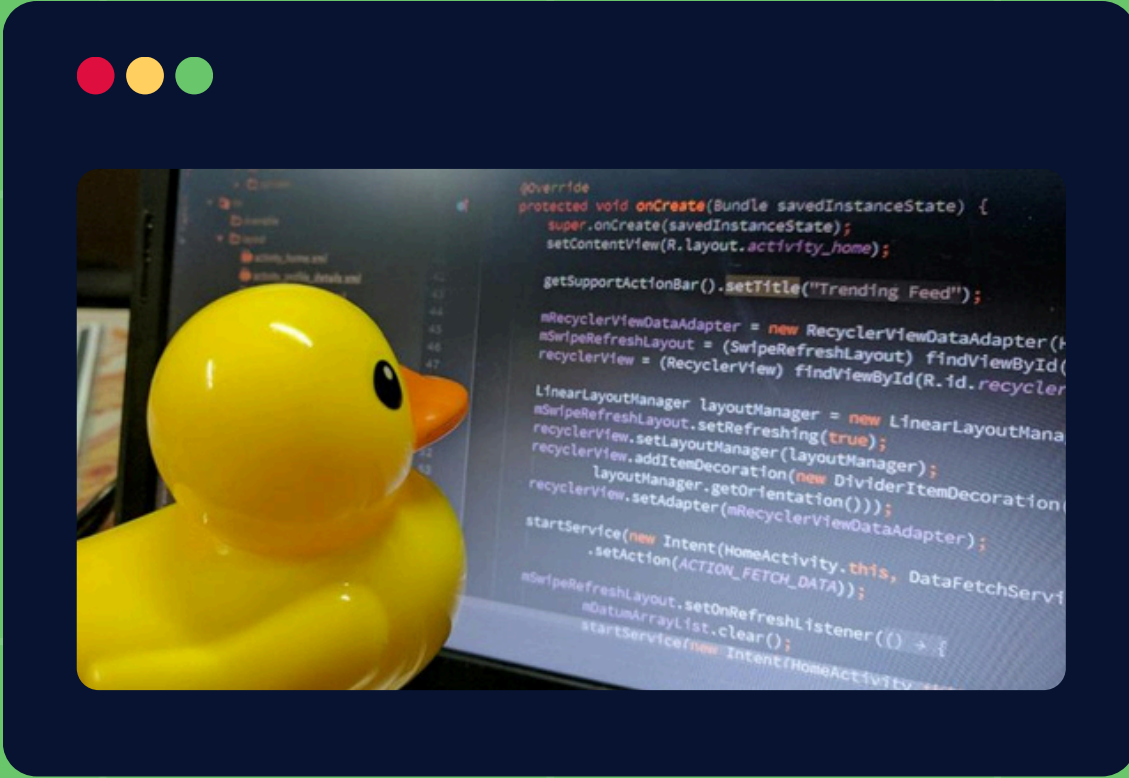




Hello World!



Support Vector Machine (SVM)

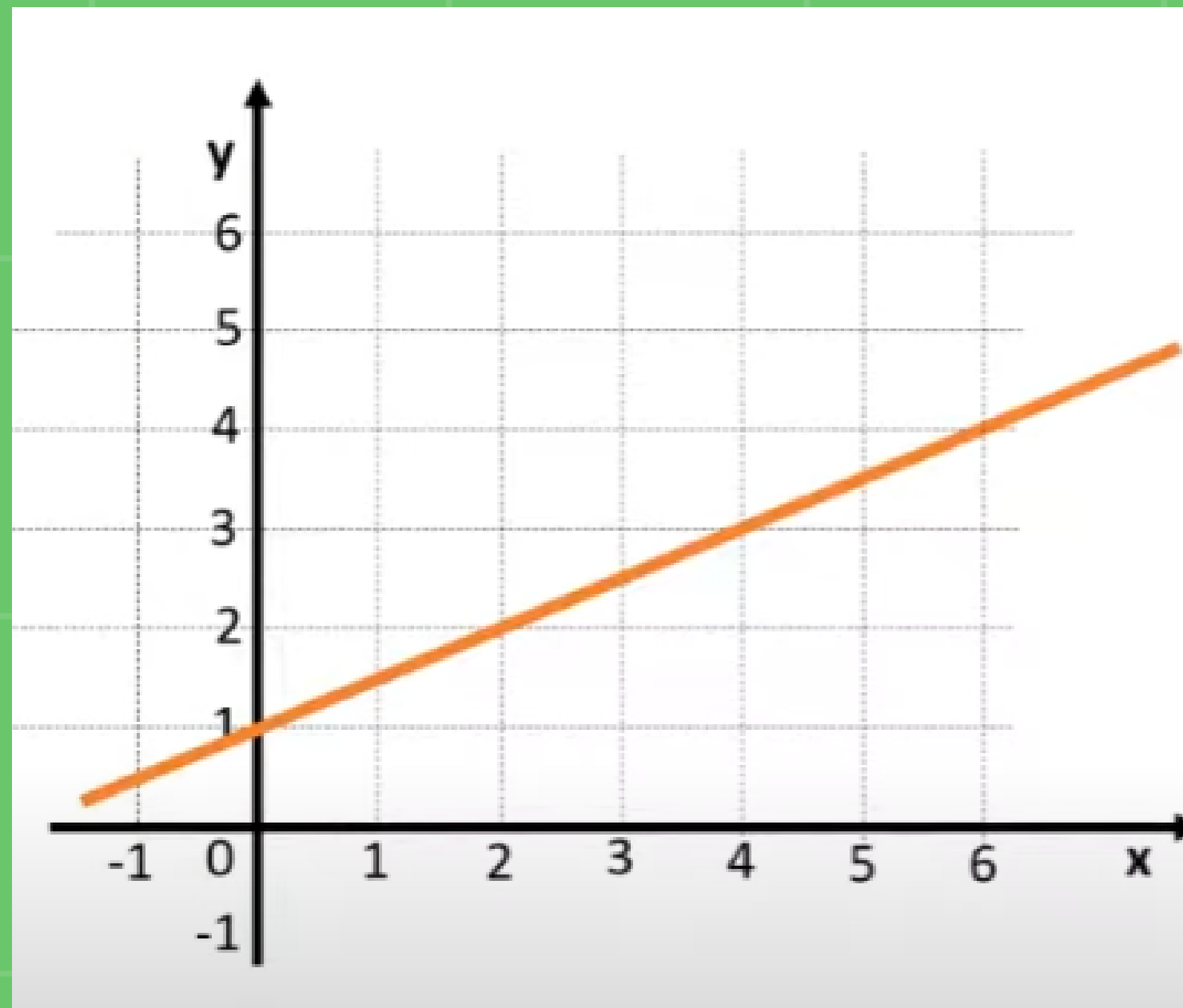


Session -> 8



Math Recap

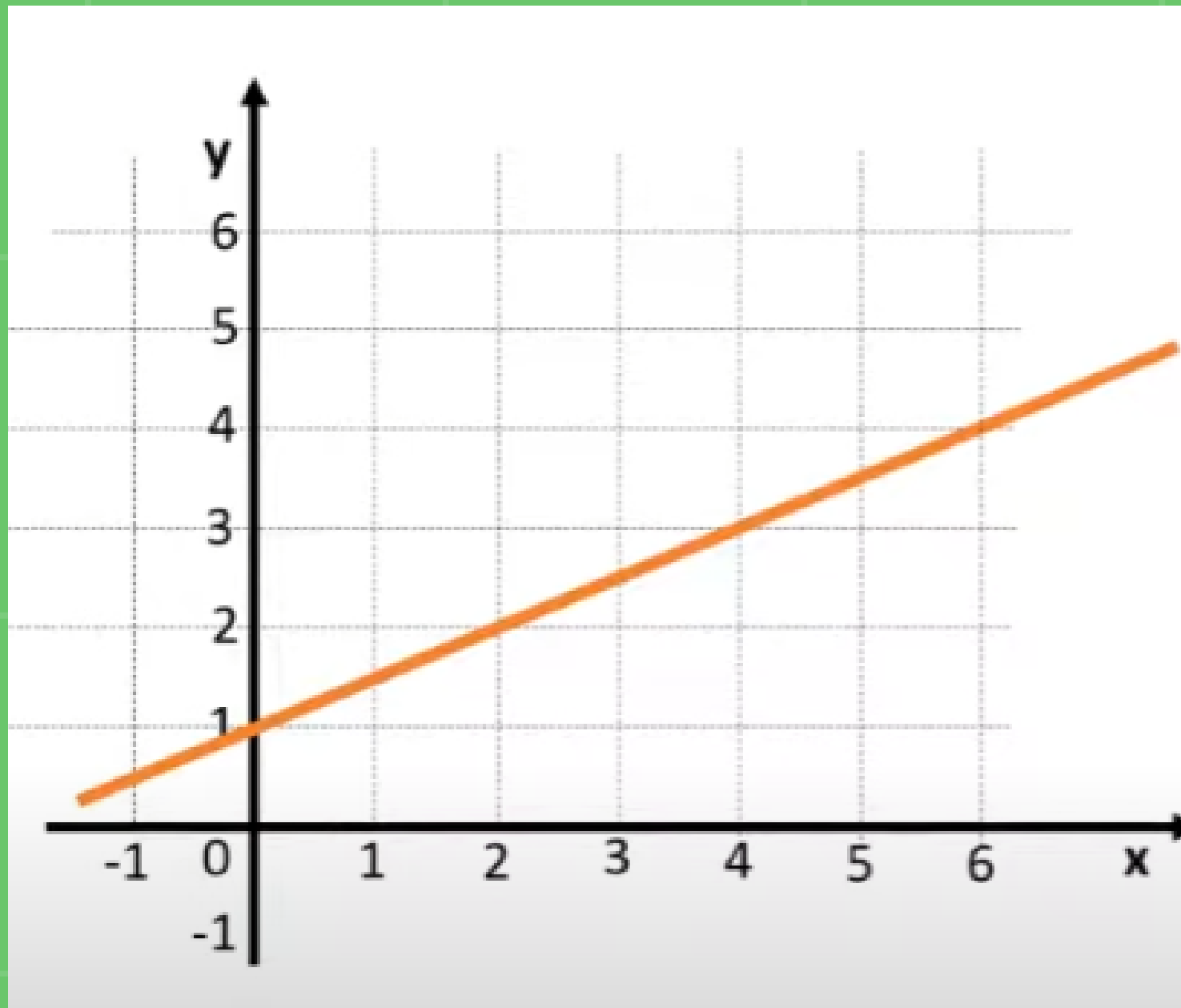
Equation of Straight Line



Remember this equation?

$$y = mx + b$$

Equation of Straight Line



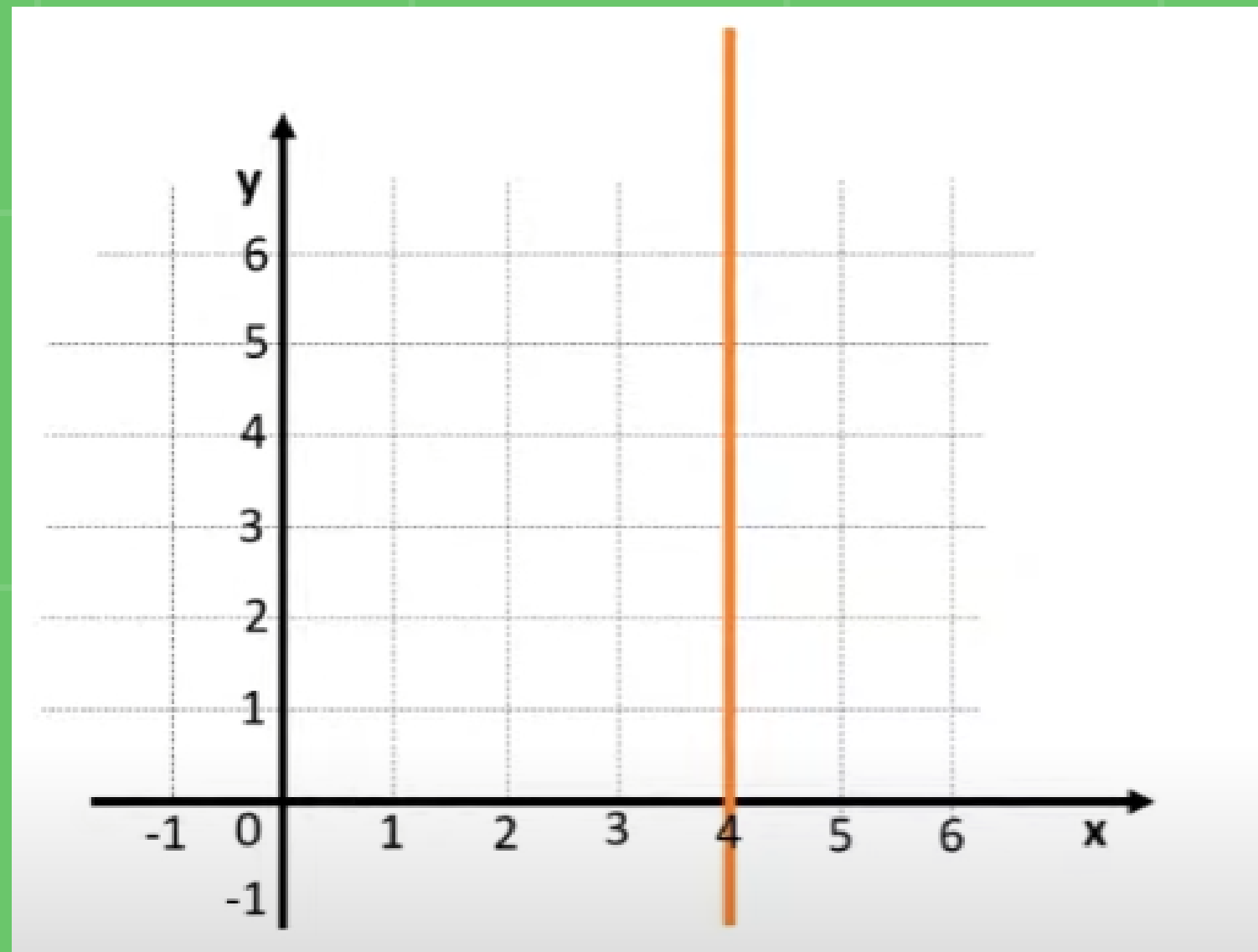
$$\rightarrow y = 0.5x + 1$$

Equation of Straight Line



$$y = 3$$

Equation of Straight Line



What about
this?

Standard Form of Line Equation

- We can express the previous line using the standard form of equation of the line which is:

$$Ax + By + C = 0$$

where A, B, and C have to be integers

Example

$$3x - 3y + 5 = 0 \longrightarrow A = 3, B = -3, C = 5$$

→ We can get the x-intercept if we set $y=0$ and vice versa

$$3x - 3(0) + 5 = 0$$

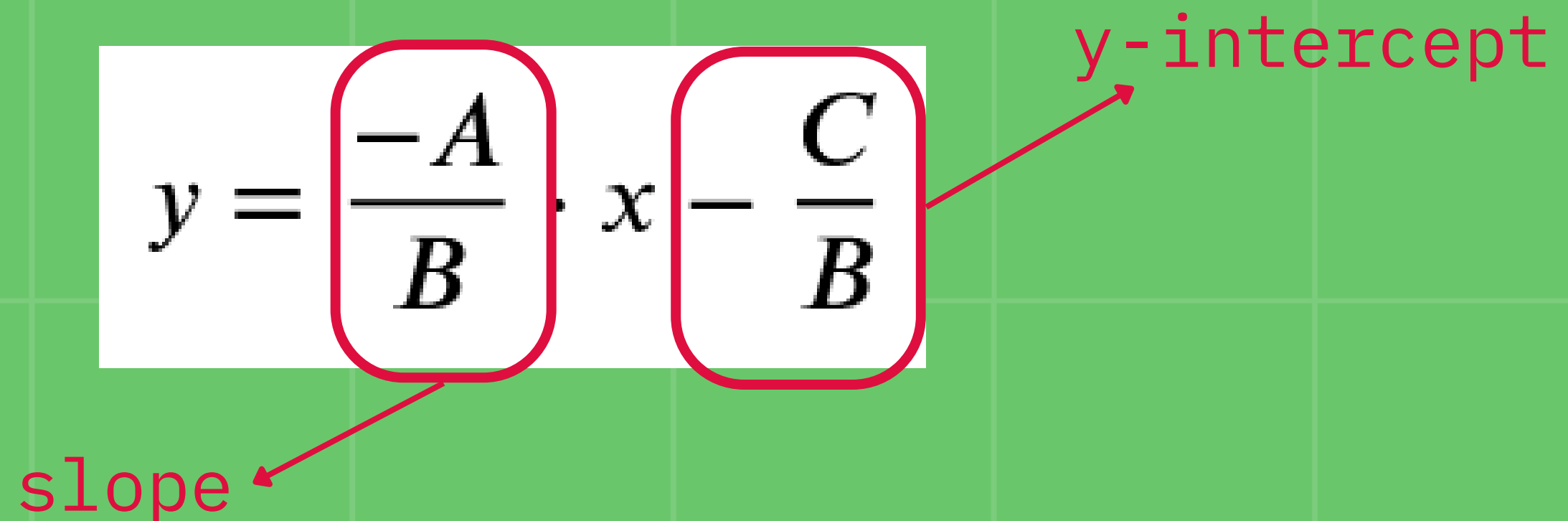
$$3x = -5$$

$$x = -1.67$$

Standard Form of Line Equation

- We can also compute the slope and the y-intercept as the following:

$$y = \frac{-A}{B} \cdot x - \frac{C}{B}$$



The diagram shows the equation $y = \frac{-A}{B} \cdot x - \frac{C}{B}$ with two red rounded rectangles highlighting the slope term $\frac{-A}{B}$ and the y-intercept term $-\frac{C}{B}$. A red arrow points from the label "slope" to the slope term, and another red arrow points from the label "y-intercept" to the y-intercept term.

$$y = \frac{-A}{B} \cdot x - \frac{C}{B}$$

slope

y-intercept

Standard Form of Line Equation

Example

$$-2x + 4y + 4 = 0 \quad (\text{standard form})$$

Then the slope: $2 / 4 = 0.5$

and the y-intercept: $-4 / 4 = 1$

$$y = 0.5x + 1 = 0 \Rightarrow -2x + 4y - 4 = 0 \quad (\text{standard form})$$

Moving a line along y-axis

- If we replaced the 0 in the standard form with 1:

$$Ax + By + C = 1$$

- Then we tried to convert $-2x + 4y - 4 = 1$ to $(y = xm + b)$

format: $-2x + 4y - 5 = 0$

slope = $-(-2/4)$, y-intercept = $-(-4/5)$

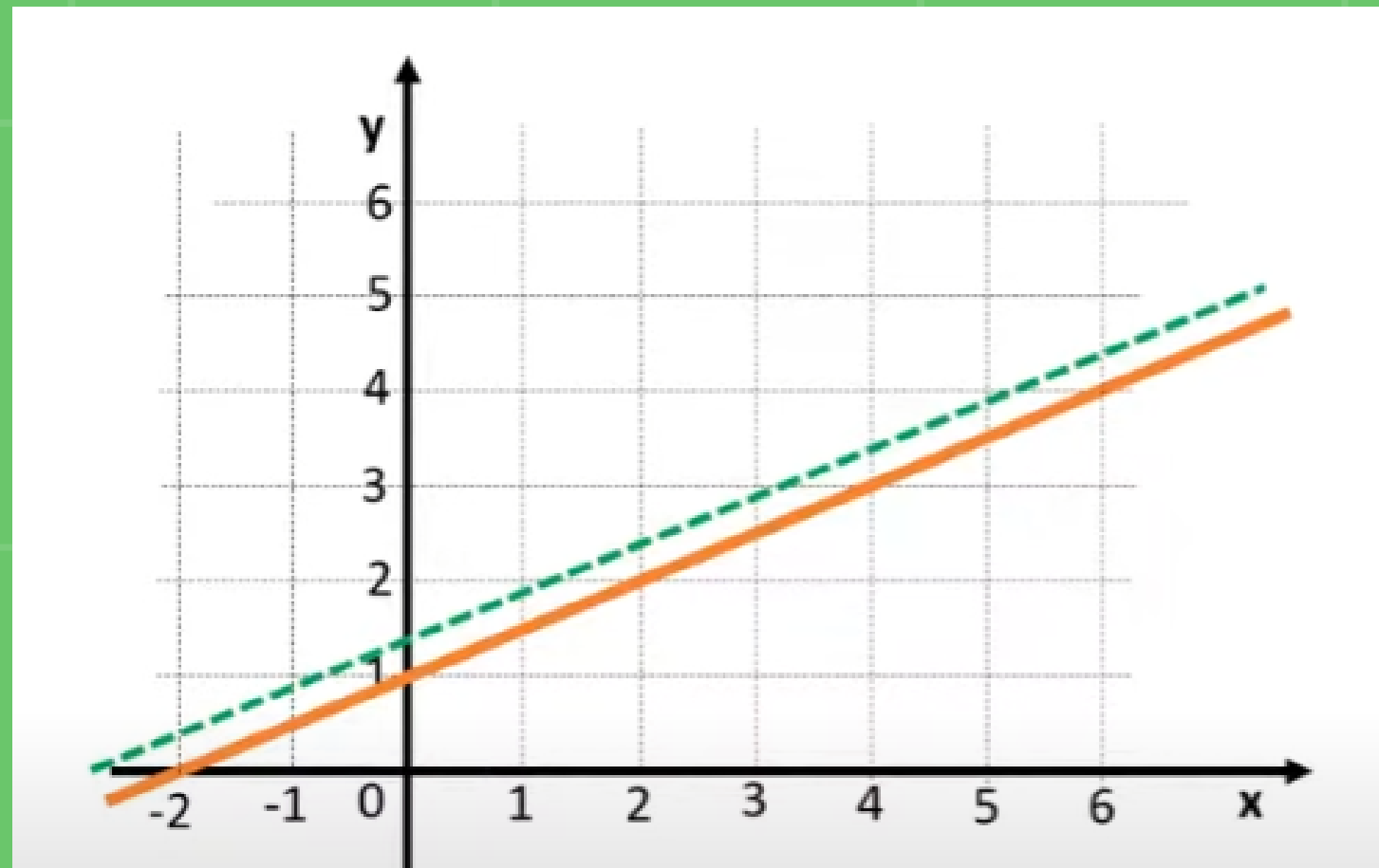
result => $y = 0.5x + 1.25$

Moving a line along y-axis

- Notice that the y-intercept **increased**, which means the new line moved **upward** but stayed parallel to the original line

$$y = 0.5x + 1 = 0 \Rightarrow -2x + 4y - 4 = 0 \quad (\text{standard form})$$

$$y = 0.5x + 1.25 = 0 \Rightarrow -2x + 4y - 4 = 1 \quad (\text{standard form})$$



Moving a line along y-axis

- If we replaced the 0 in the standard form with -1:

$$Ax + By + C = -1$$

- Then we tried to convert $-2x + 4y - 4 = -1$ to $(y = xm + b)$

format: $-2x + 4y - 3 = 0$

slope = $-(-2/4)$, y-intercept = $-(-4/3)$

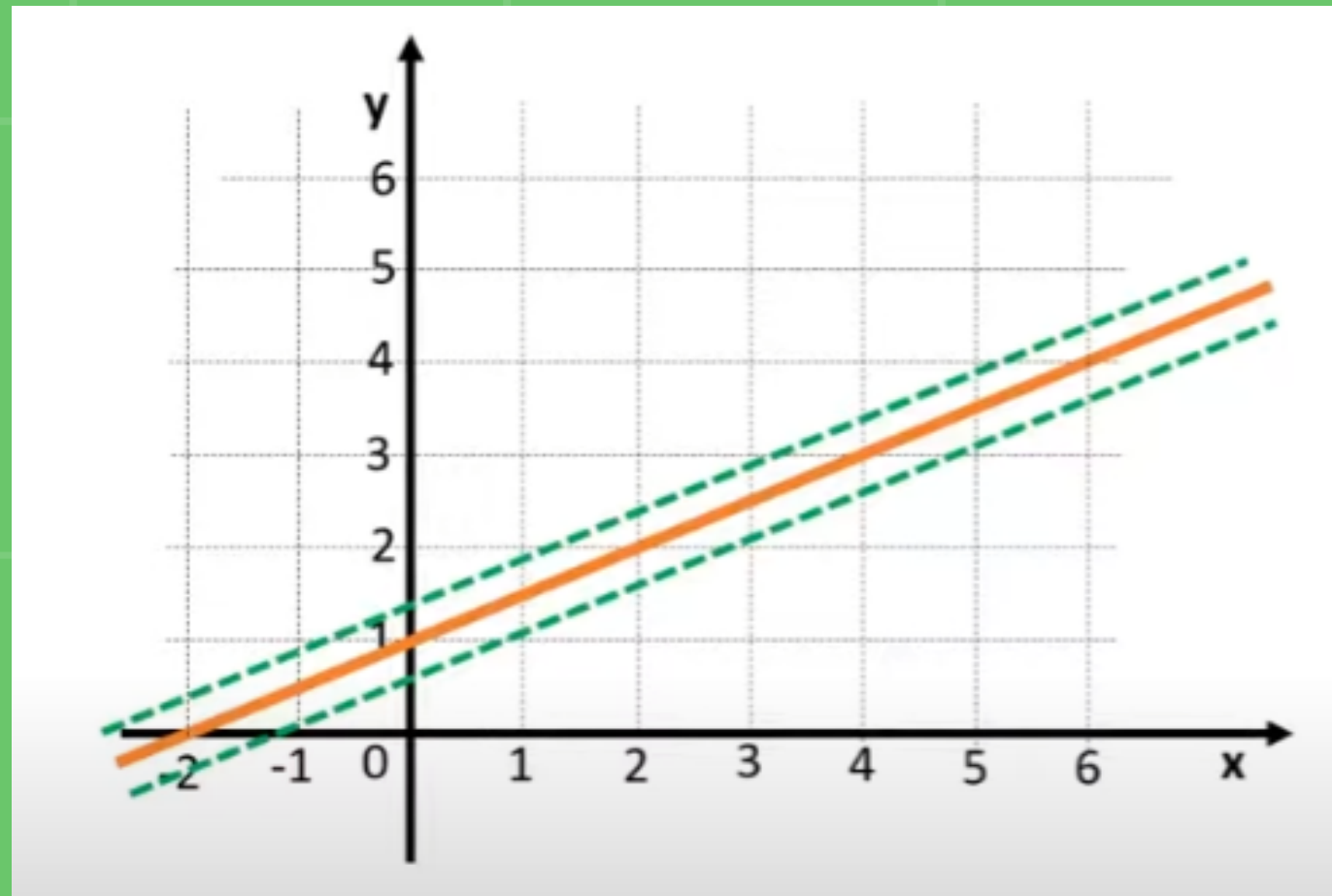
result => $y = 0.5x + 0.75$

Moving a line along y-axis

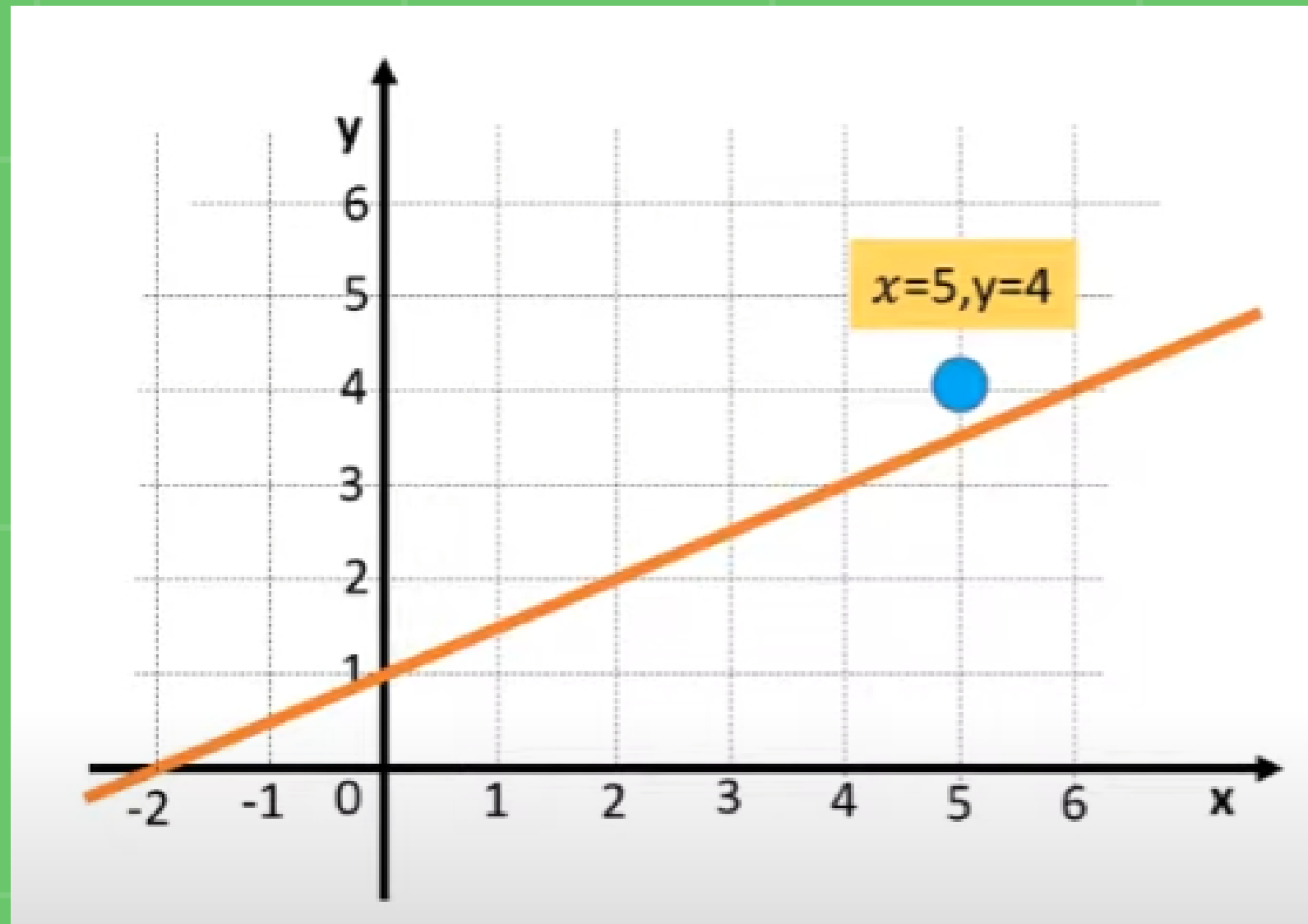
- Notice that the y-intercept **decreased**, which means the new line moved **downward** but stayed parallel to the original line

$$y = 0.5x + 1 = 0 \Rightarrow -2x + 4y - 4 = 0 \quad (\text{standard form})$$

$$y = 0.5x + 0.75 \Rightarrow -2x + 4y - 4 = -1 \quad (\text{standard form})$$



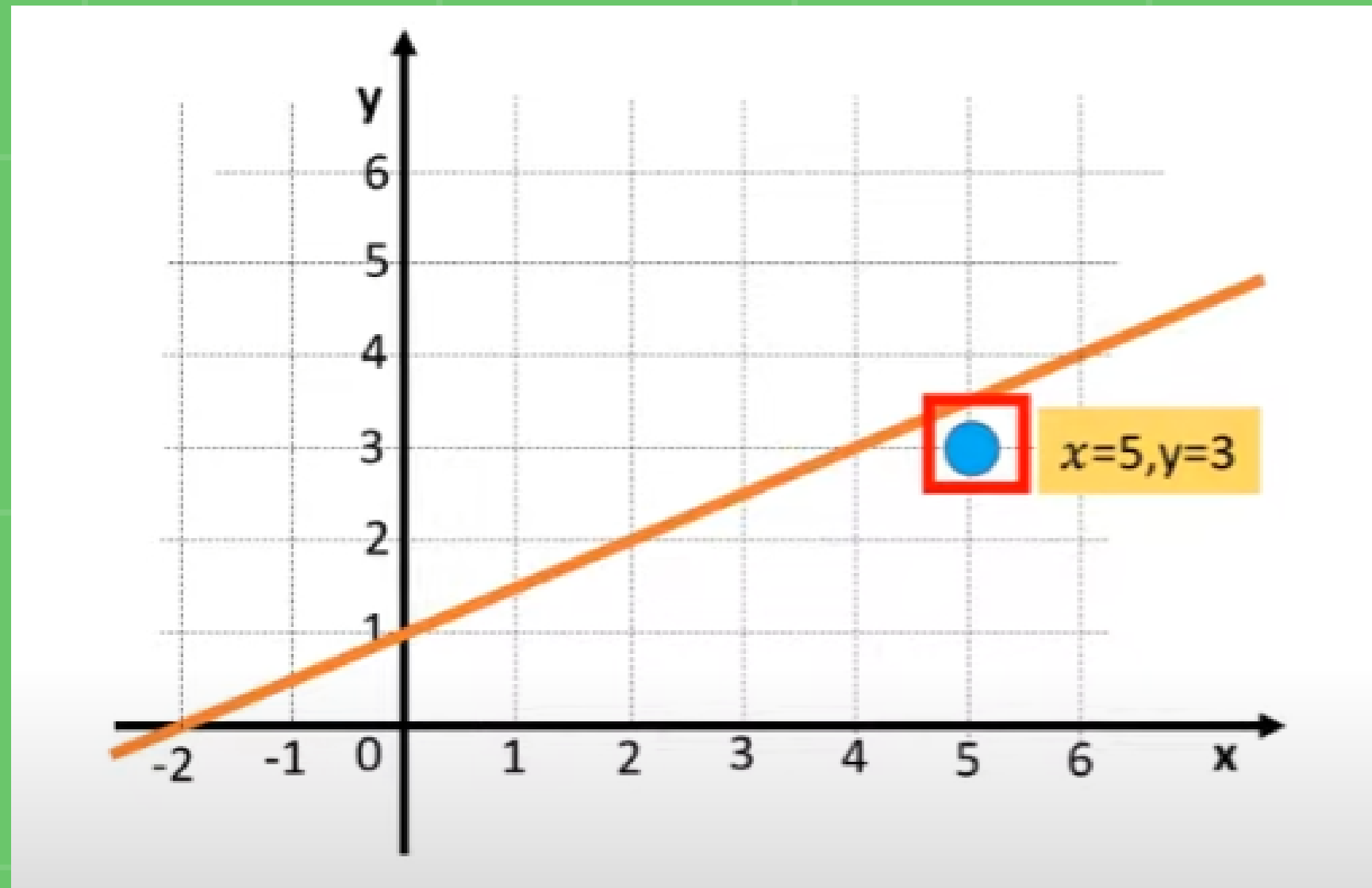
Determine whether a point is above or below a line



- Say we have a point $d1$
 $d1 = (5, 4)$
- The standard form is:
 $-2x + 4y - 4 = 0$
 $-2(5) + 4(4) - 4 = 2$

Then $d1$ is above the line

Determine whether a point is above or below a line



- Say we have a point d1
d2 = (5,3)
- The standard form is:
 $-2x + 4y - 4 = 0$
 $-2(5) + 4(3) - 4 = -2$

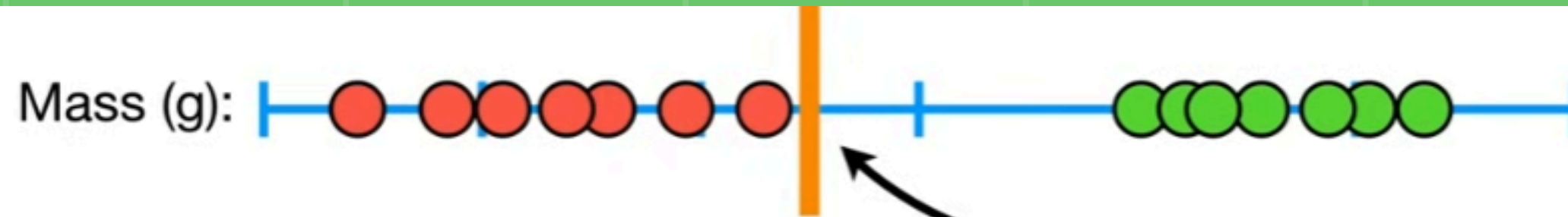
Then d2 is below the line



Intro to SVM

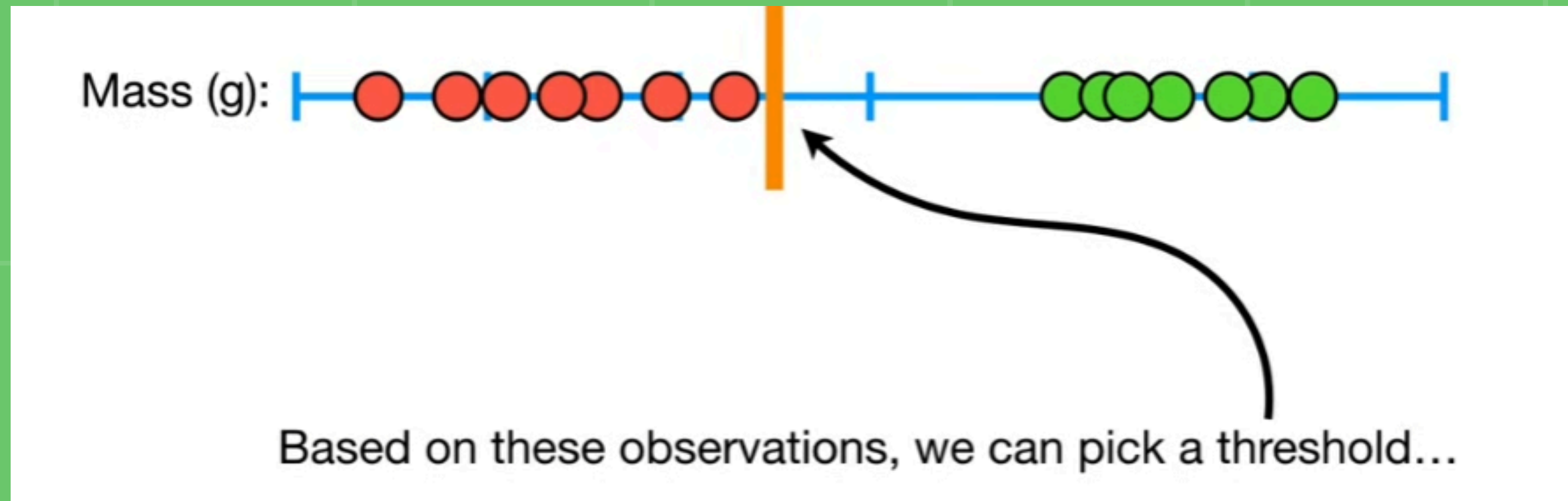
Classifying 1D dataset

- We have a dataset that contains two classes = (obese, not obese) based only on the mass



Based on these observations, we can pick a threshold...

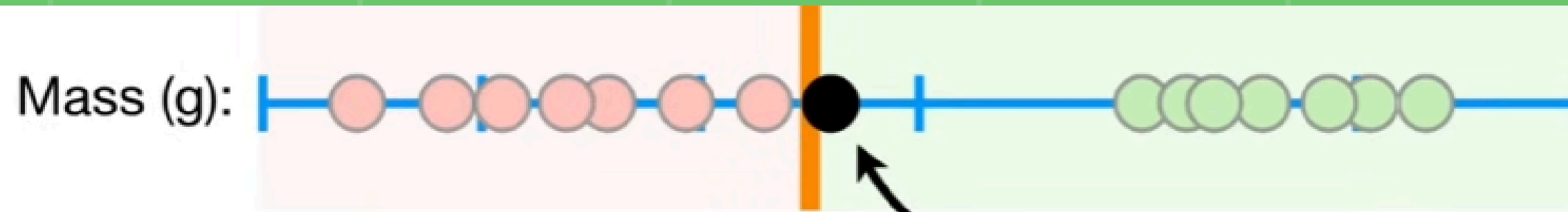
Classifying 1D dataset



If we have a new datapoint that has a mass below the threshold,
Then it is **not obese**

If we have a new datapoint that has a mass above the threshold,
Then it is **obese**

Classifying 1D dataset

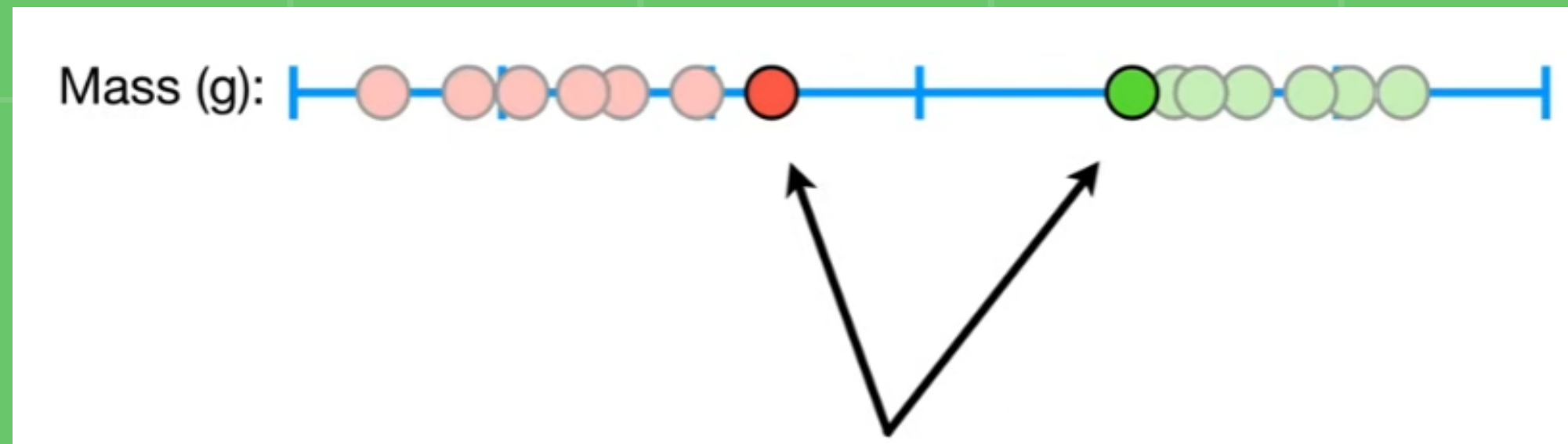


However, what if get a new observation here?

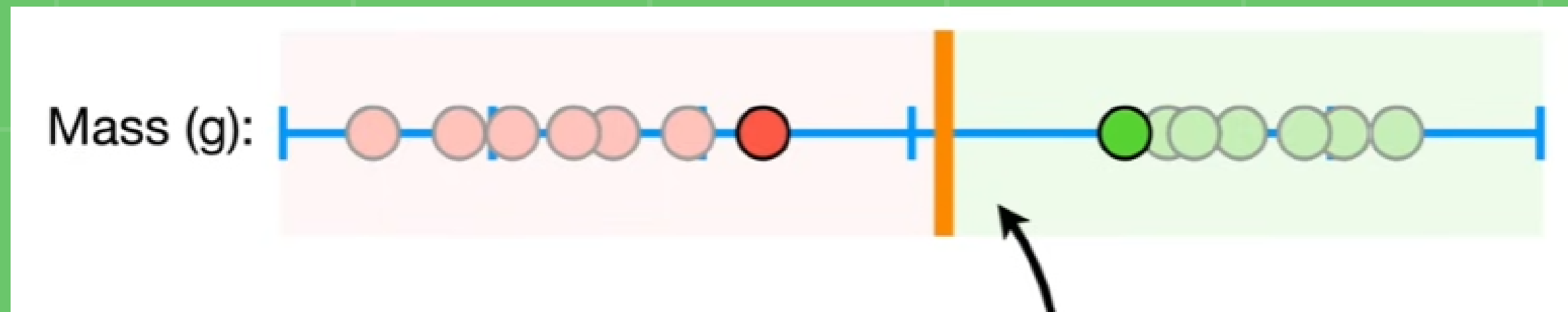
It will be classified as **obese** because it is above the threshold, although it is closer to not obese..

Classifying 1D dataset

- So we will choose a new threshold in the middle based on the points in the edges for each class

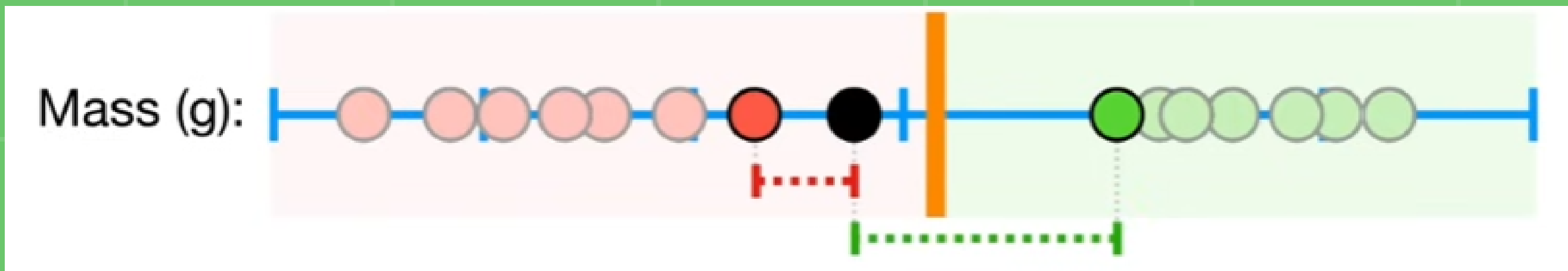


New threshold:

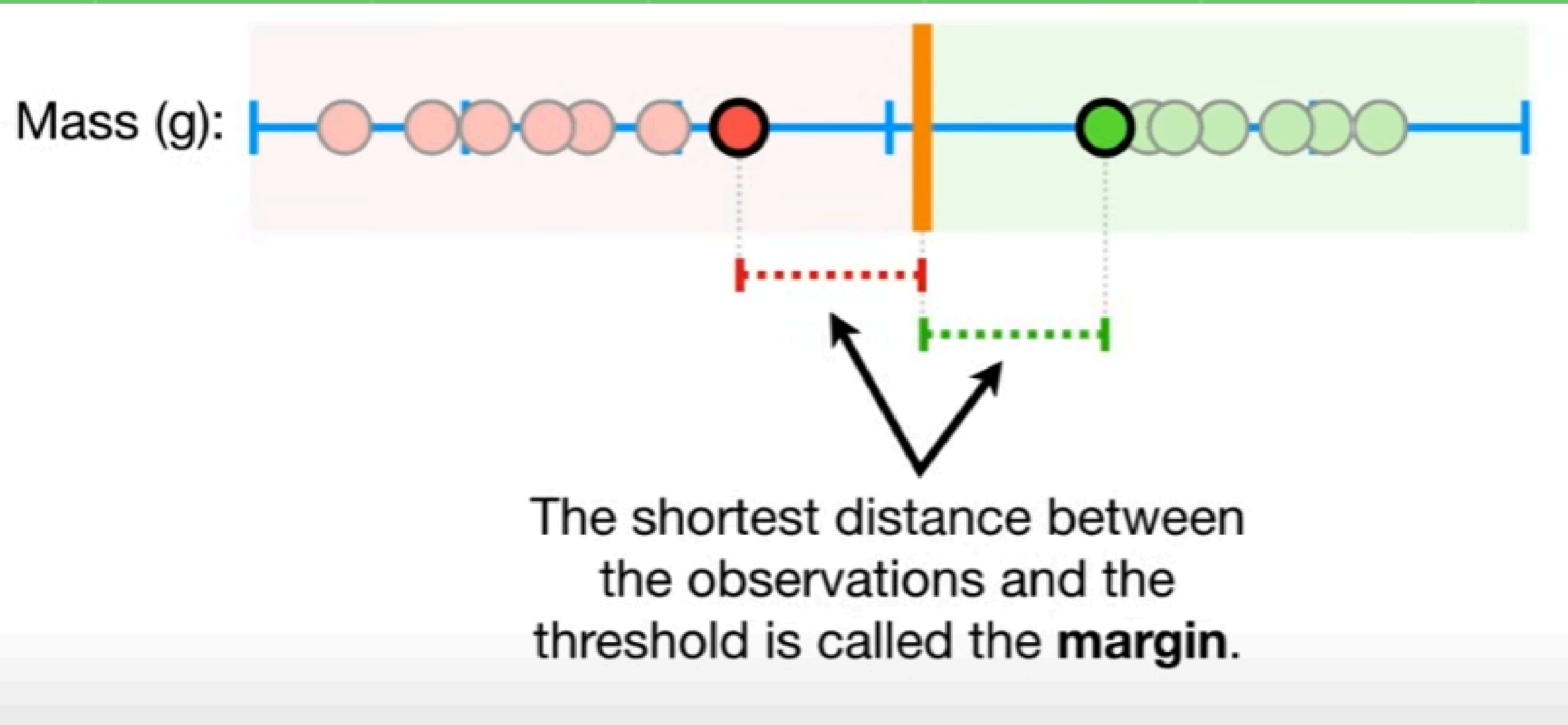


Classifying 1D dataset

- Now, if we have the same new datapoint on the left side of the threshold, it will be classified as **not obese**, because it is closer to the edge of the 'not obese' class:



Classifying 1D dataset



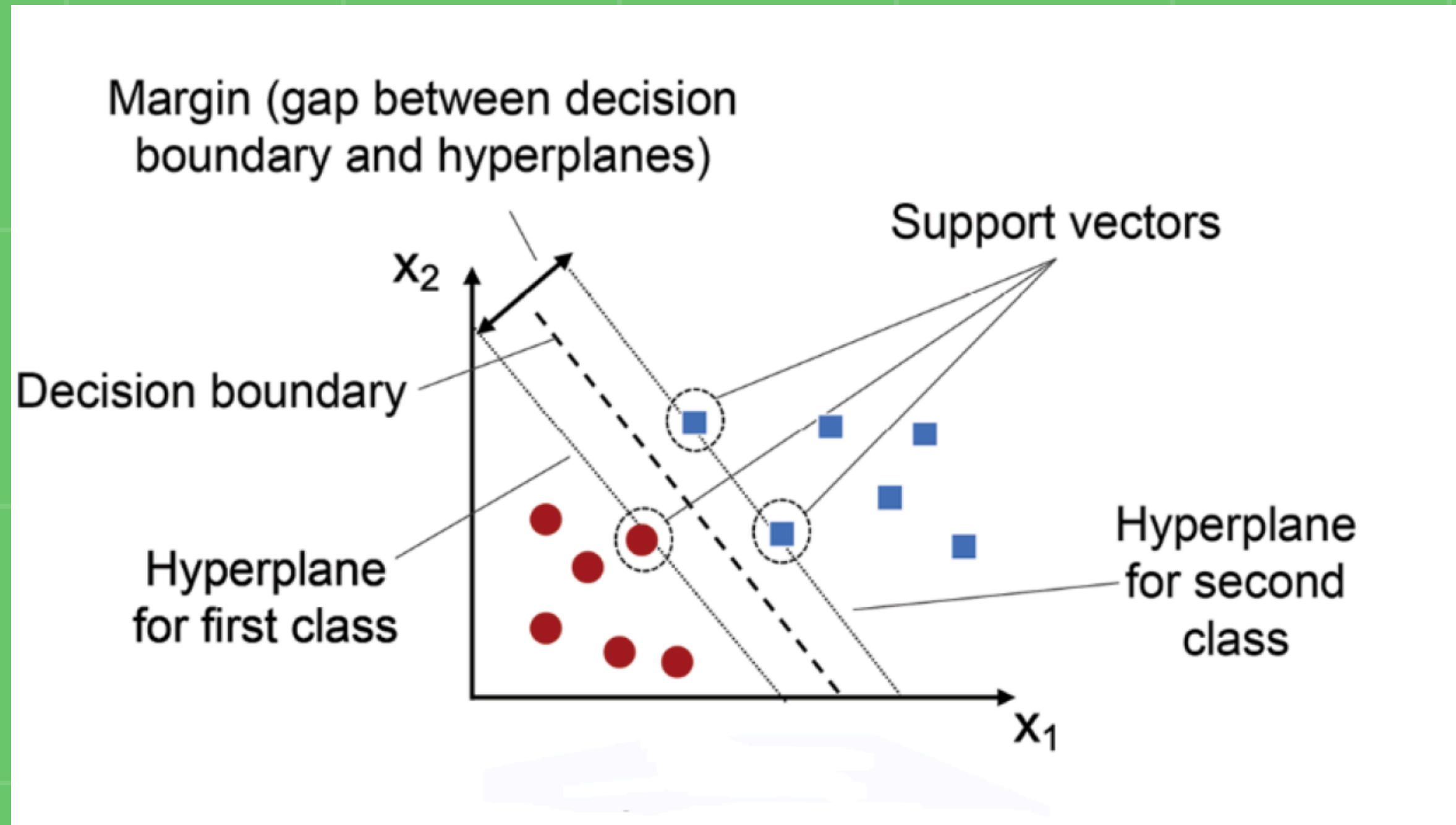


Support Vector Machines

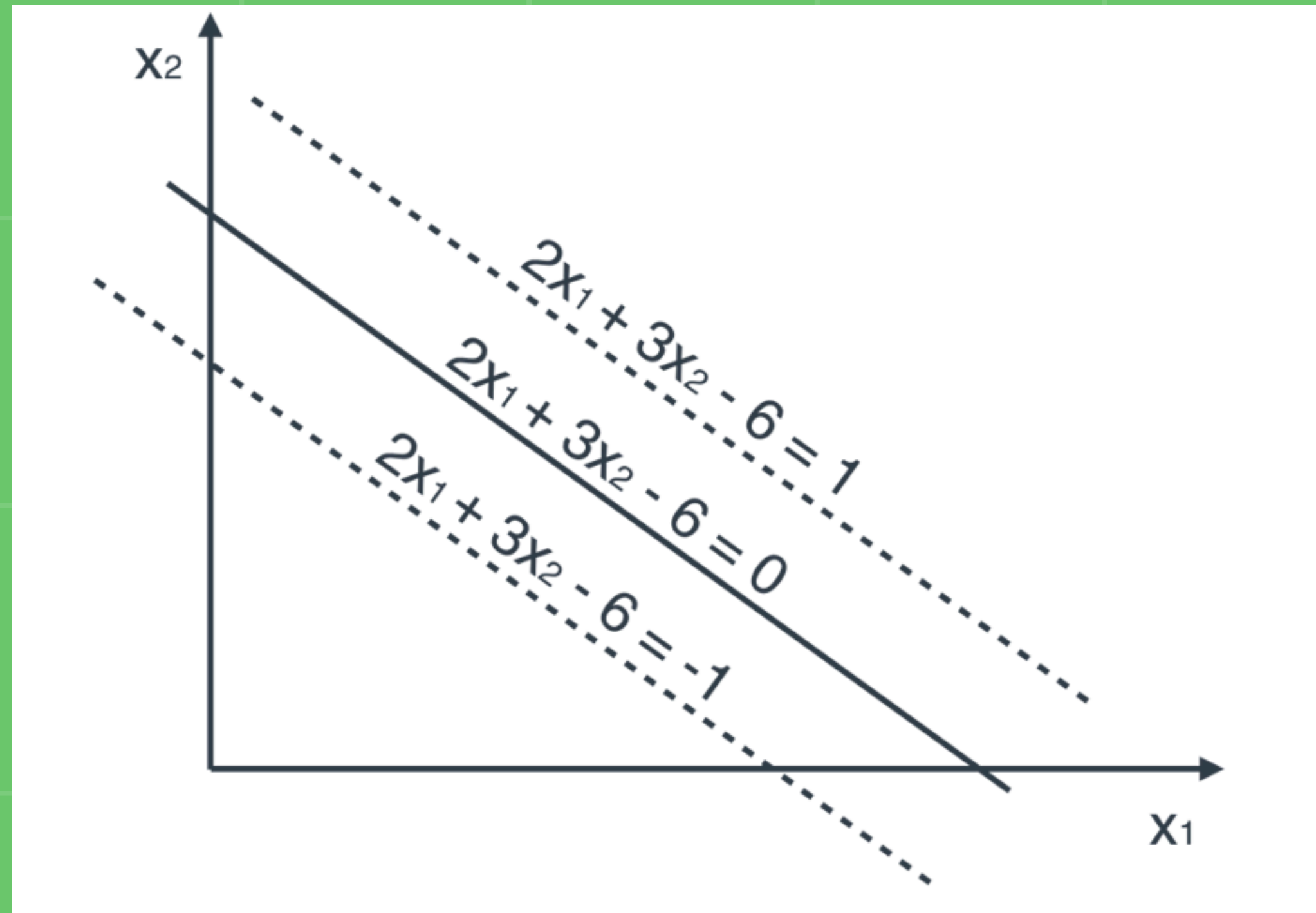
What is SVM?

- Support Vector Machine (SVM) is a **supervised, discriminative** machine learning algorithm that is most commonly used for **classification** tasks.
- The primary goal of SVM is to find the optimal decision boundary (hyperplane) that separates classes in the feature space with the **maximum possible margin**, helping the model generalize well to unseen data.

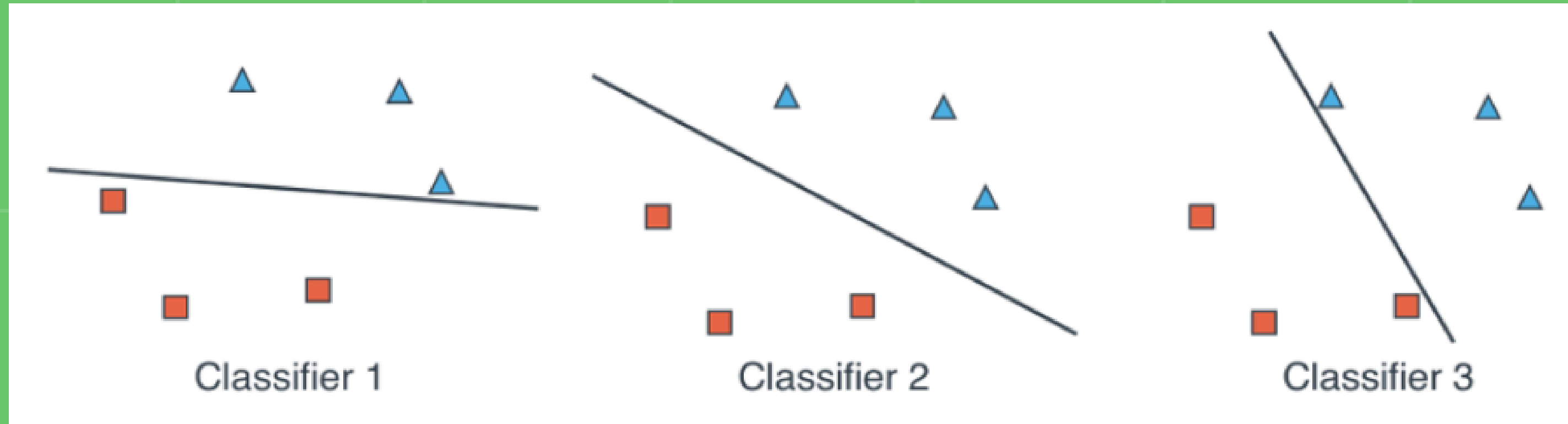
What is SVM?



What is SVM?

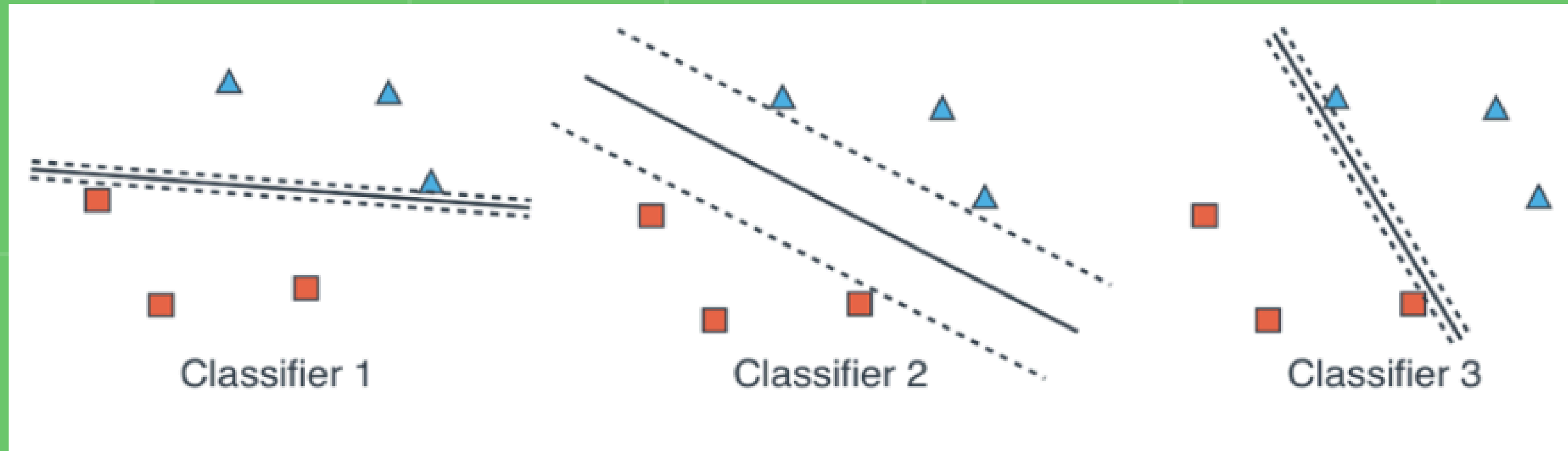


Which classifier is best?



- The 3 separate the data well, but the **2nd** classifier is the best, because the line is far from the points, where the 1st and the 3rd are very close to some, which may lead to misclassification.

Which classifier is best?



- SVM classifier uses two parallel lines instead of one line.
- Goal of SVM: classify the data correctly and maximize the margin as possible.



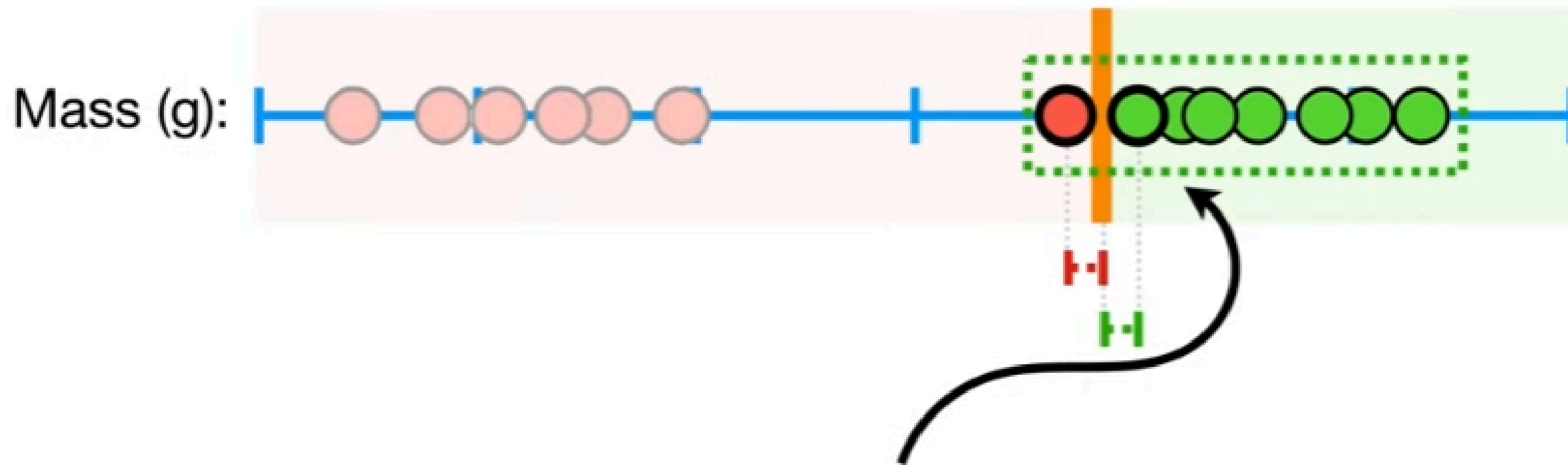
Hard Margin **vs** Soft Margin

Classifying obese dataset



- Remember the obese dataset? Imagine we have a dataset like this, the **'not obese'** class contains outliers that are close to the second class.

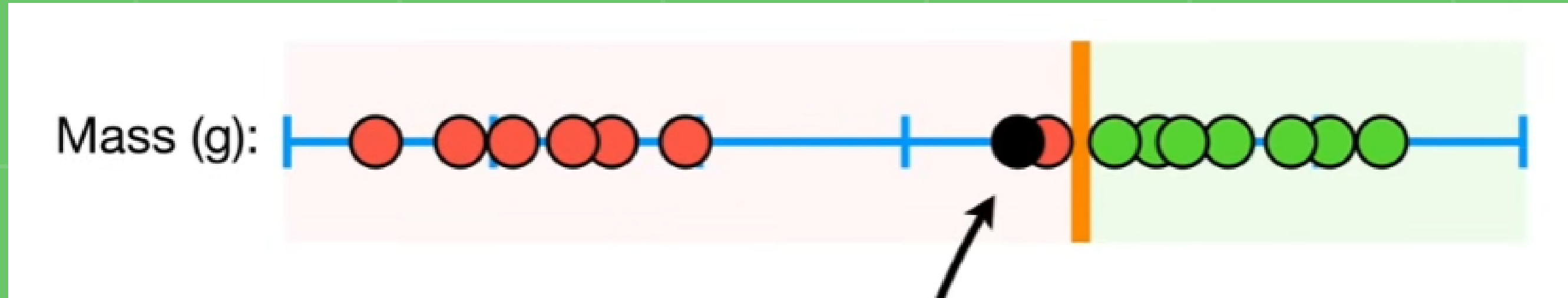
Classifying obese dataset



In this case, the **Maximum Margin Classifier** would be super close to the **obese** observations...

- The boundary will be affected by the outlier and move to the right.

Classifying obese dataset

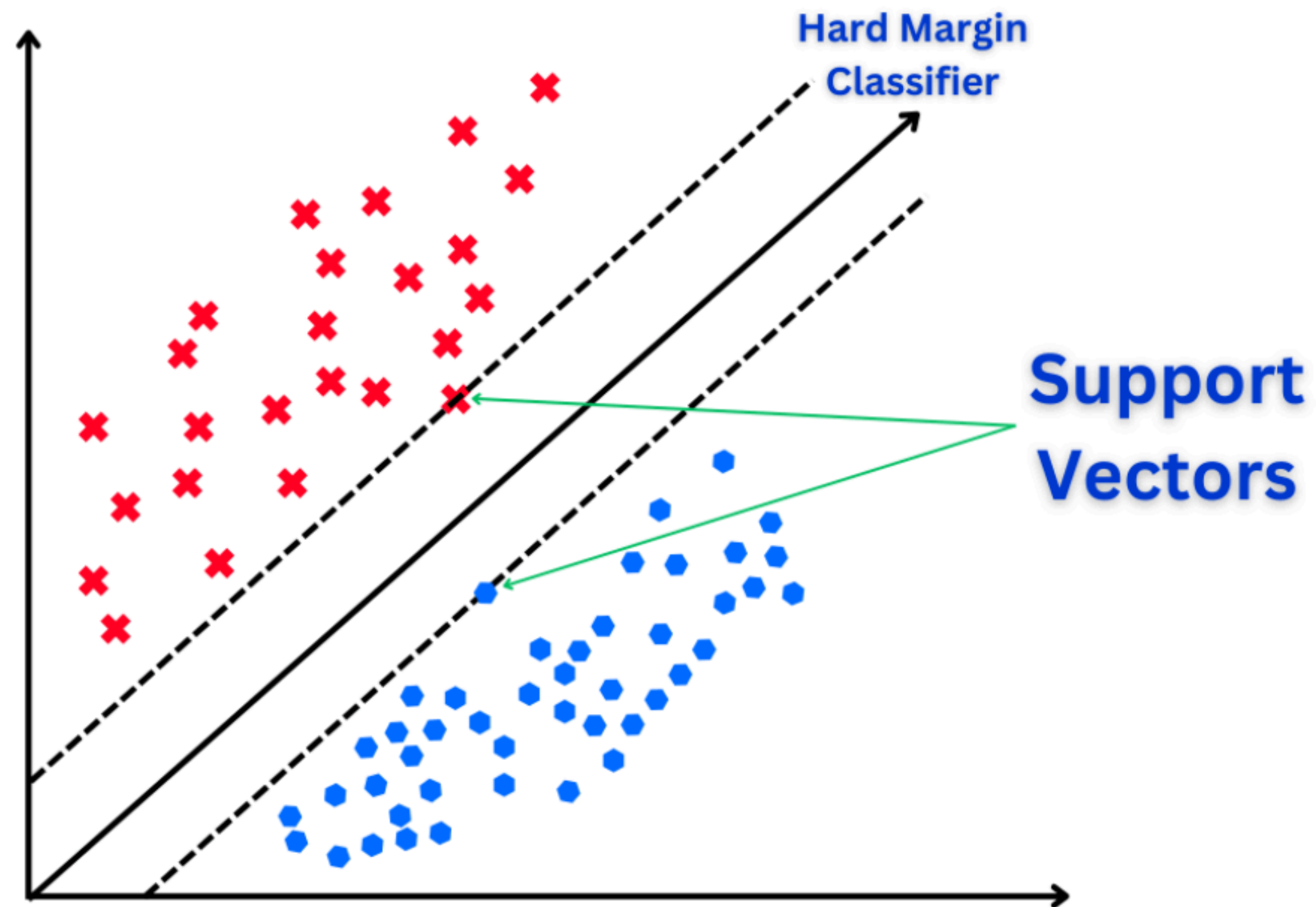


- Now, if we have a new datapoint that is likely to be obese, it will be misclassified as **not obese**, because the boundary is far away from the first class
- We can now see how the decision boundry is **sensitive to outliers**.

Hard Margin in SVM

- A hard margin SVM requires that the data be **perfectly linearly separable**, meaning there should be a clear, unbroken boundary between classes without any overlap or misclassified points.
- In a hard margin setup, SVM aims to **maximize the margin as much as possible**, ensuring that no data points fall within the margin or on the wrong side of the hyperplane.

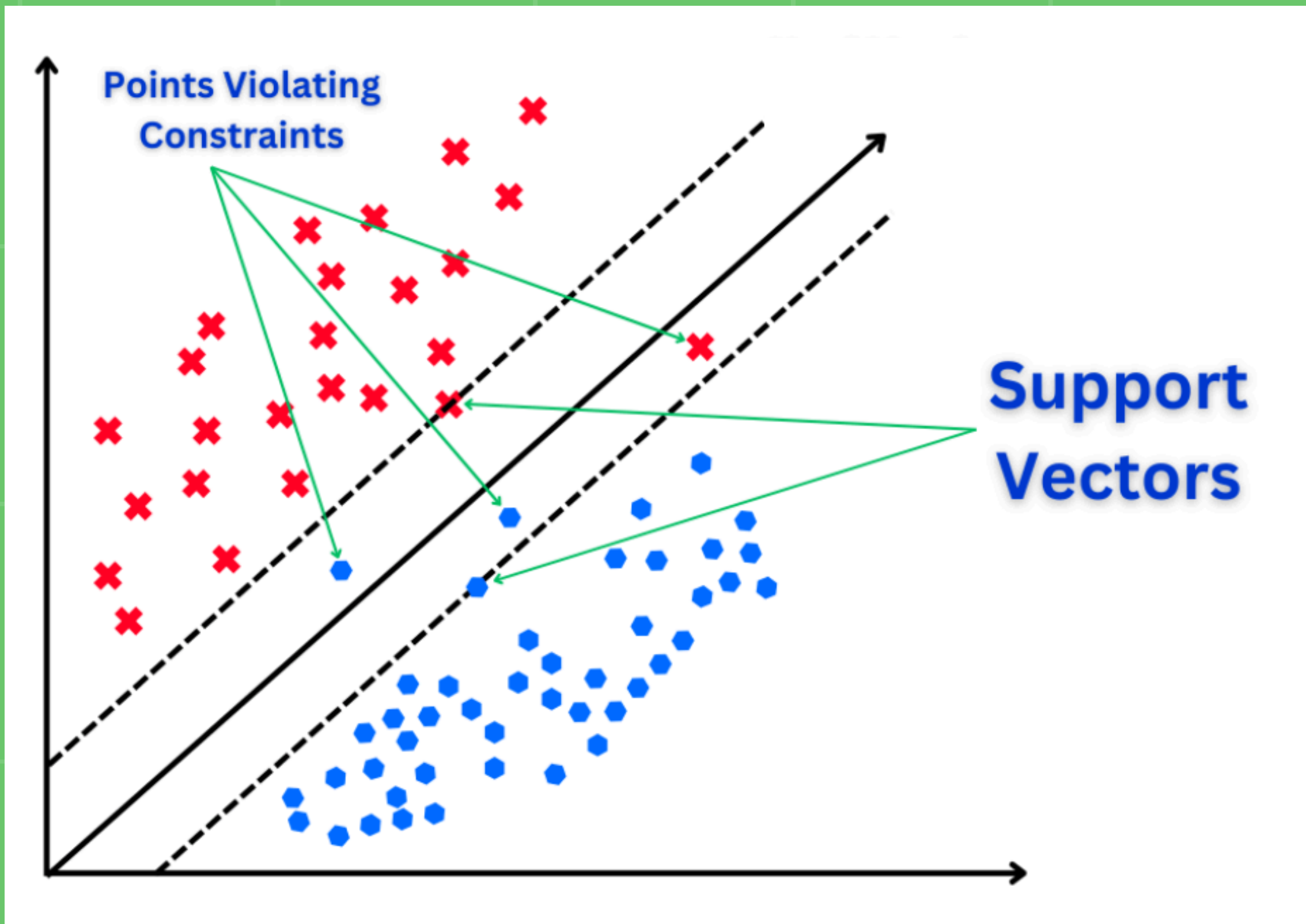
Hard Margin in SVM



Soft Margin in SVM

- A Soft Margin SVM tries to make the margin between classes **as wide as possible**, while still **allowing some mistakes**. It uses special values (called **C parameter**) to let a few points be inside the margin or even on the wrong side of the line. This makes the model more flexible and better at handling real-world data that isn't perfectly separable.

Soft Margin in SVM





Hinge Loss

Hinge loss function

- To compute the cost in SVM, we use 'Hinge loss function'

$$\text{Hinge Loss} = \max(0, 1 - y \cdot f(x))$$

- $\mathbf{f(x)} = \mathbf{w^t x + b}$: the predicted score
- $\mathbf{y}$ = the true label (-1, 1)

Hinge loss function

$$\text{Hinge Loss} = \max(0, 1 - y \cdot f(x))$$

- **If $y \cdot f(x) \geq 1$** → Correct and outside the margin → **no loss.**
- **If $0 < y \cdot f(x) < 1$** → Correct but inside the margin → **some loss.**
- **If $y \cdot f(x) \leq 0$** → Misclassified → **larger loss.**

Hinge loss function

Example: Let's say that $y = +1$

- If $f(x) = 2$:

Hinge Loss = $\max(0, 1 - (1 * 2)) = \max(0, -1) = 0$ (perfect)

- If $f(x) = 0.5$:

Hinge Loss = $\max(0, 1 - (1 * 0.5)) = \max(0, 0.5) = 0.5$ (inside margin)

- If $f(x) = -1$:

Hinge Loss = $\max(0, 1 - (1 * -1)) = \max(0, 2) = 2$ (misclassified)



Error Function

Error Function in SVM

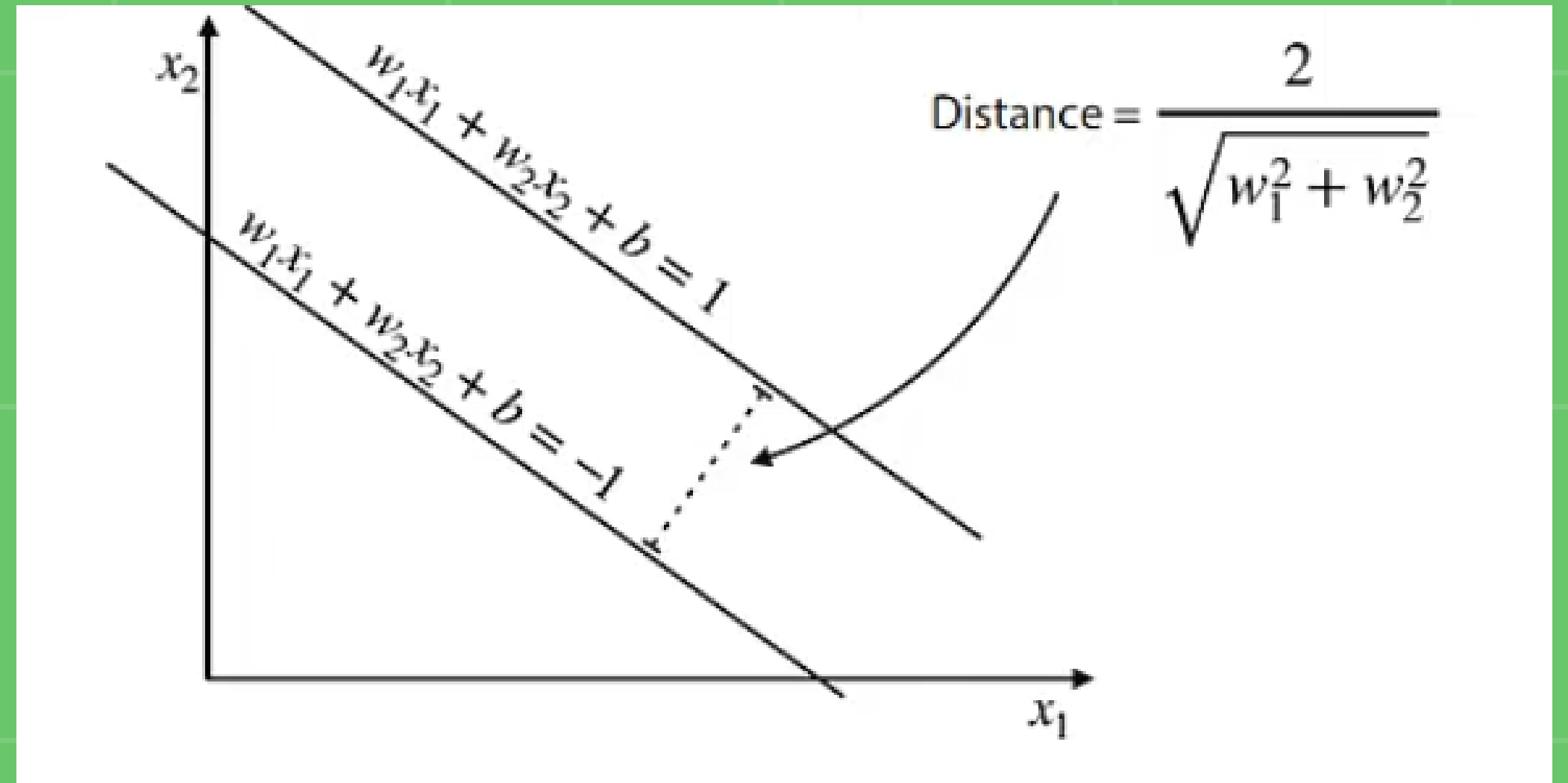
- It tries to maximize two things at the same time, the true classifications, and the distance between the classes.

Error = Classification error + Distance Error

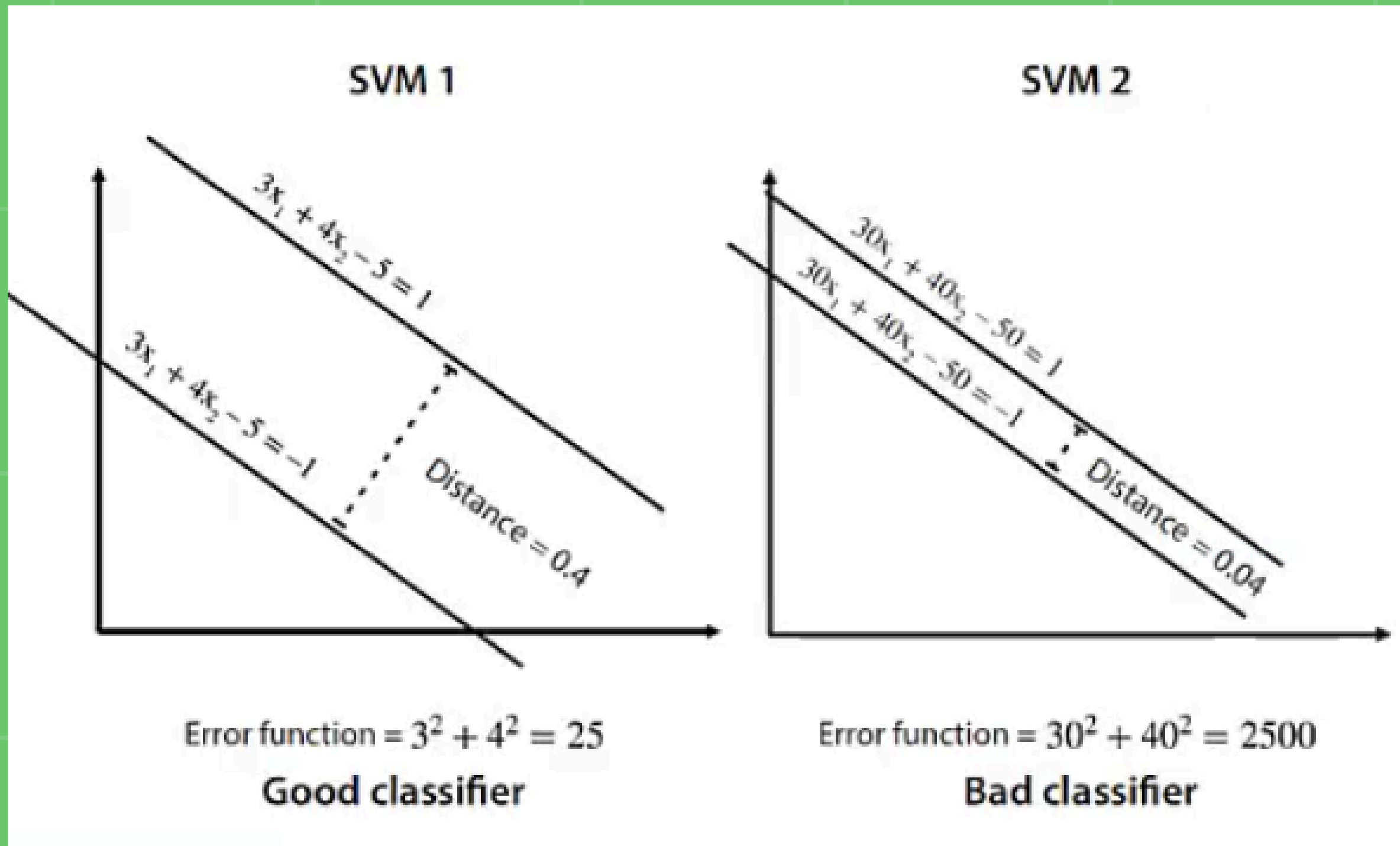
Distance Error

- The distance between two lines is calculated using this formula:

$$\frac{2}{\sqrt{w_1^2 + w_2^2}}$$

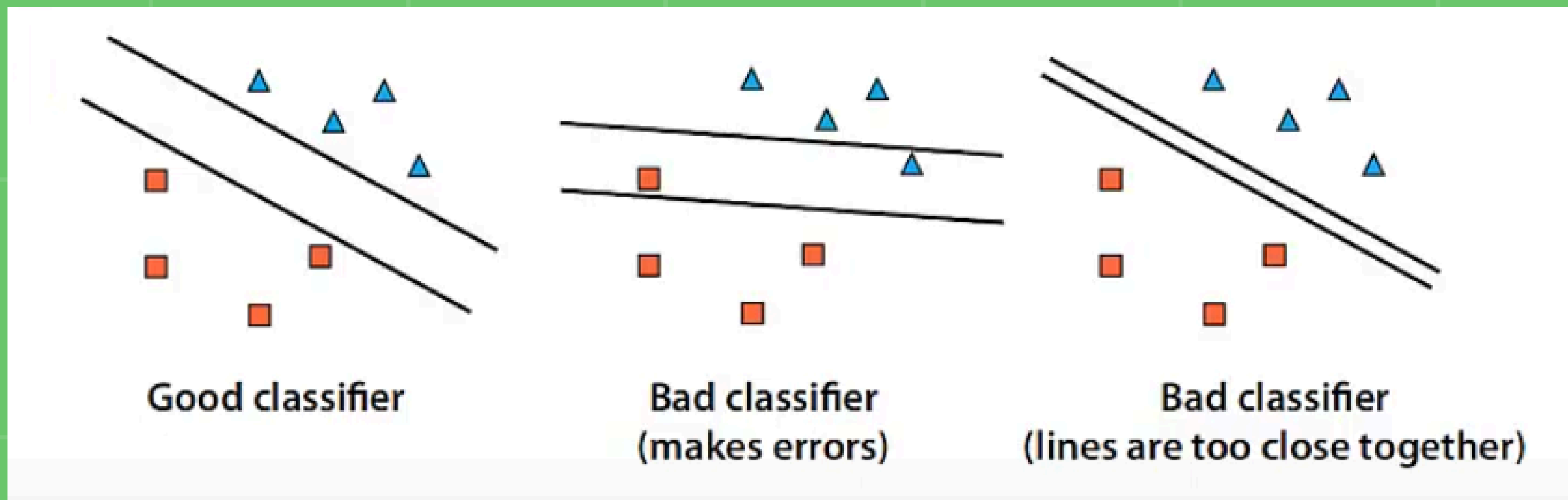


Distance Error



Error Function in SVM

- A good SVM with low error tries to make as few classification errors as possible, while trying to keep the lines as far apart as possible.

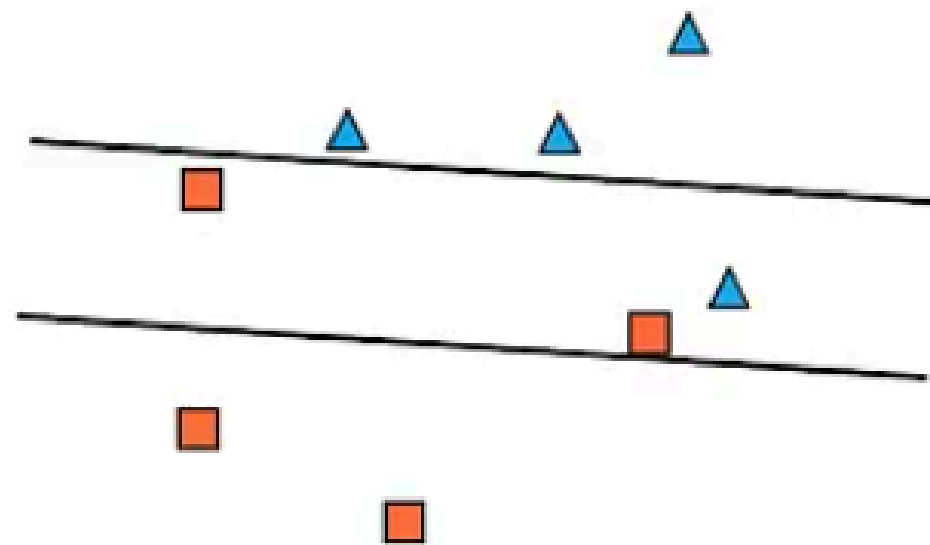


Error Function in SVM

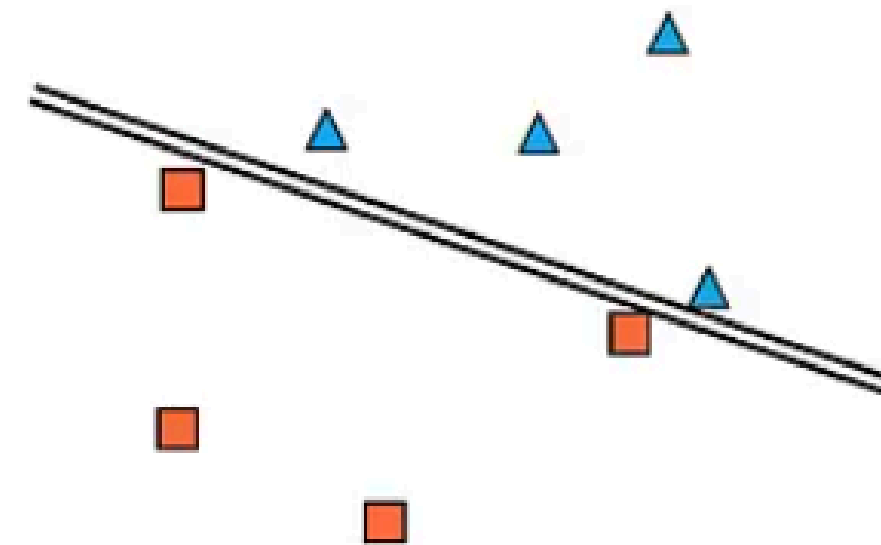
$$\min_{\mathbf{w}, b, \xi_i} \quad \frac{1}{2} ||\mathbf{w}'||^2 + C \sum_{i=1}^n \xi_i$$

Error Function in SVM

- But what if we have to sacrifice one of distance and cost error for the benefit of the other?
- Which one should we prefer?



Lines are far apart (good).
Makes errors (bad).



Lines are too close (bad).
Doesn't make errors (good).

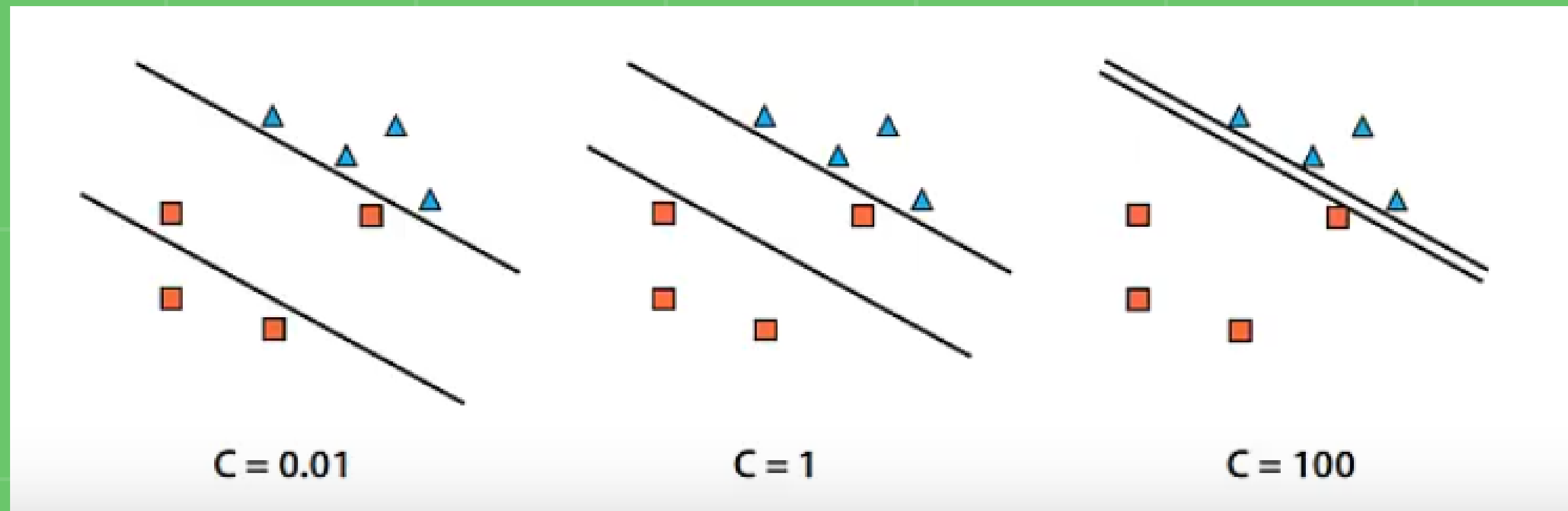
C parameter

$$\text{Error} = \text{C} * \text{Classification error} + \text{Distance Error}$$

- If C is large, then the classification error is dominant, so our classifier focuses more on classifying the points correctly.
- If C is small, then the distance error is dominant, so our classifier focuses more on maximizing the distance between the two lines.

C parameter

$$\text{Error} = \text{C} * \text{Classification error} + \text{Distance Error}$$



Hard margin & C parameter

- In hard-margin SVM, we do not allow any margin violations or misclassifications. That's equivalent to saying:
 - Set C parameter to a very large number.
 - Violations are heavily penalized.
 - Only works if the data is perfectly linearly separable.

Soft margin & C parameter

- In real-world data, perfect separation is rare or leads to overfitting. That's where soft-margin SVM comes in:
 - Use balanced C to allow some misclassifications.
 - Model ignores some misclassified points to get a larger margin, reducing overfitting risk



Kernel Trick

Dosage dataset

- We have a new dataset of the number of people who was cured based on the dosage of the drug

Dosage (mg): 

→ Notice that if the dosage is too low or too high, people are not cured (class 0), but if they take a balanced dosage, they are cured (class 1).

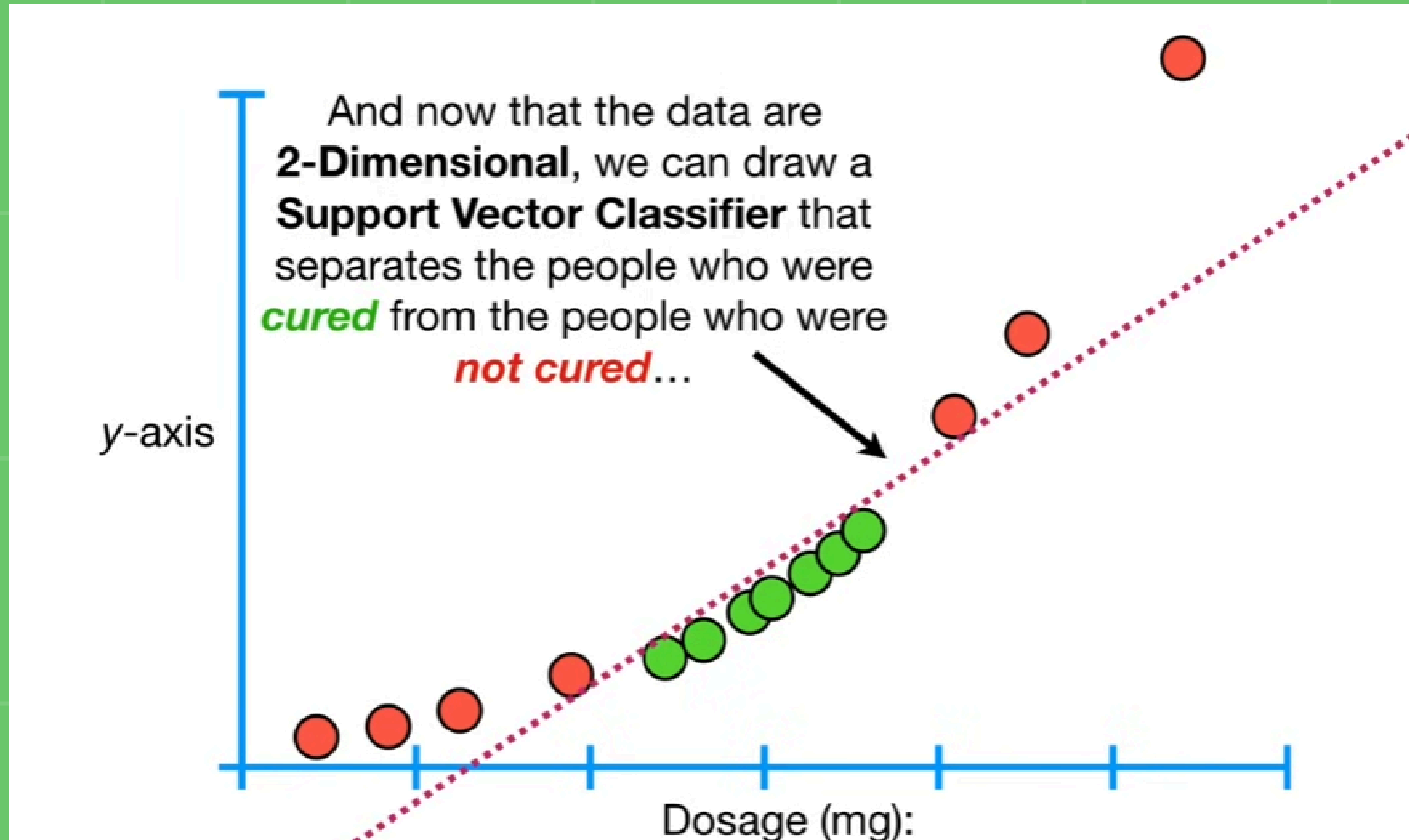
How Can We draw a boundary?

- In 1D, there is no way to put a good boundary, that's why we have the kernel.

Dosage (mg): 

Let's try to make it 2D by putting the $(\text{dosage})^2$ on the y-axis

How Can We draw a boundary?



What is the kernel trick?

- The Kernel Trick allows SVM (or other algorithms) to operate in a high-dimensional feature space without ever computing the coordinates of that space explicitly.

Types of kernels:

- Linear Kernel
- Polynomial Kernel
- Radial Basis Function (RBF)
- Sigmoid Kernel

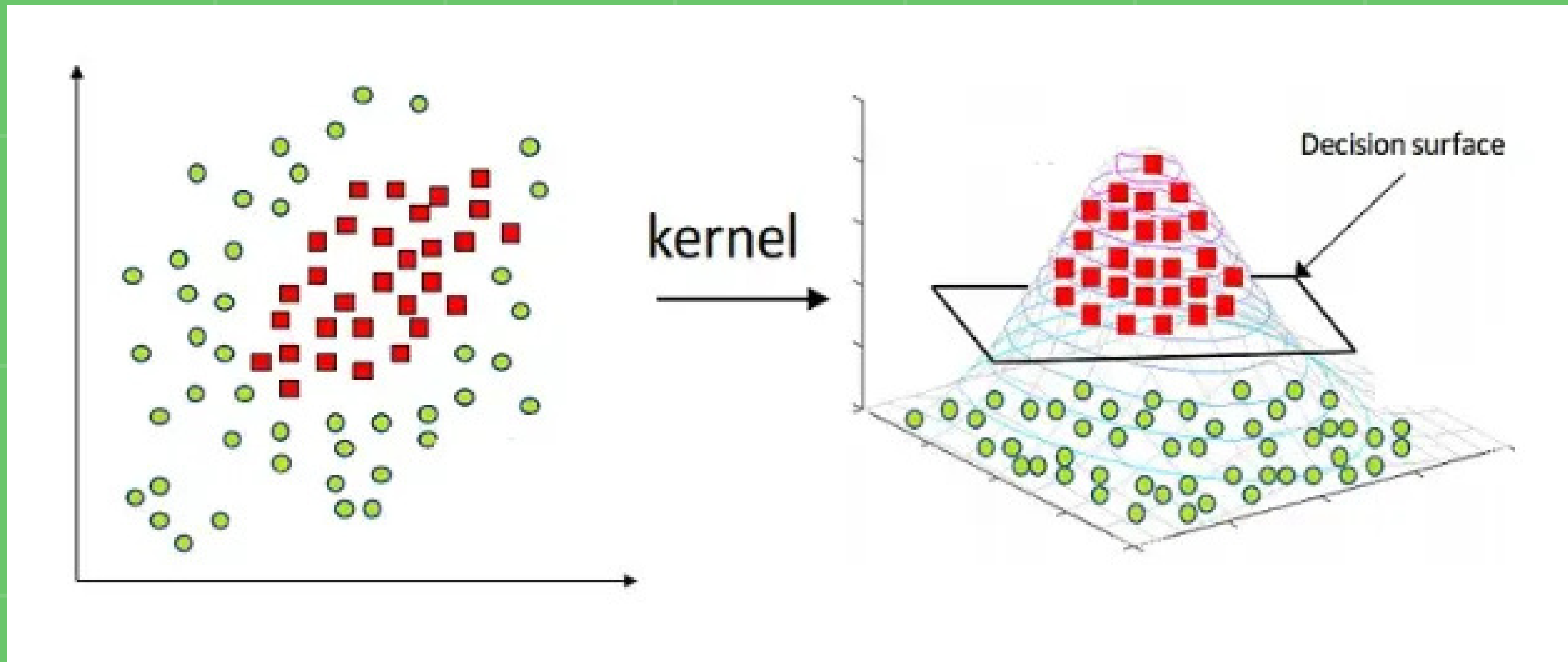
Polynomial Kernel

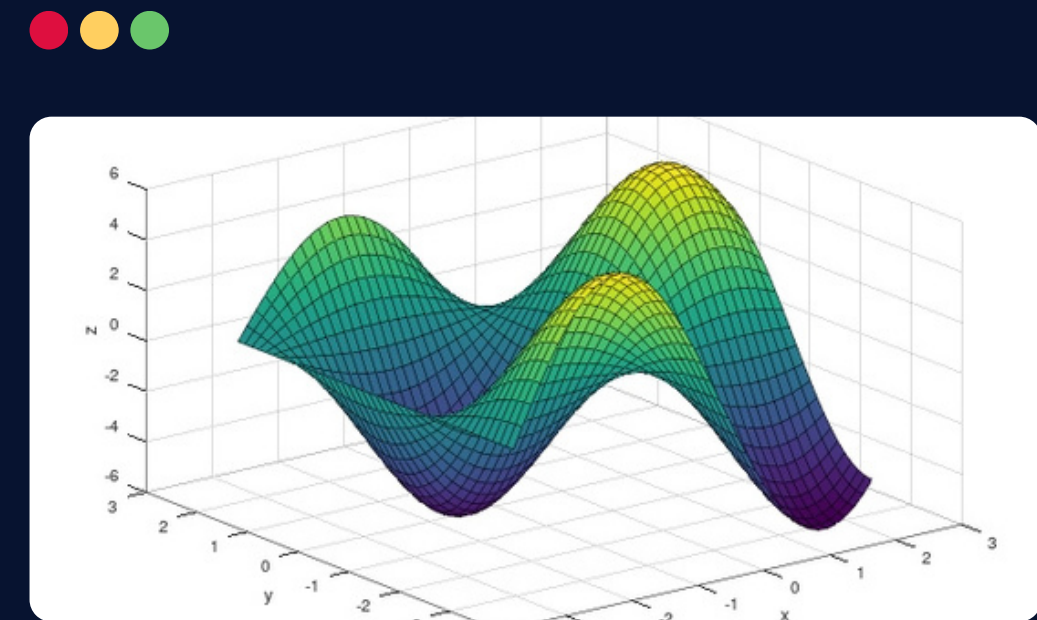
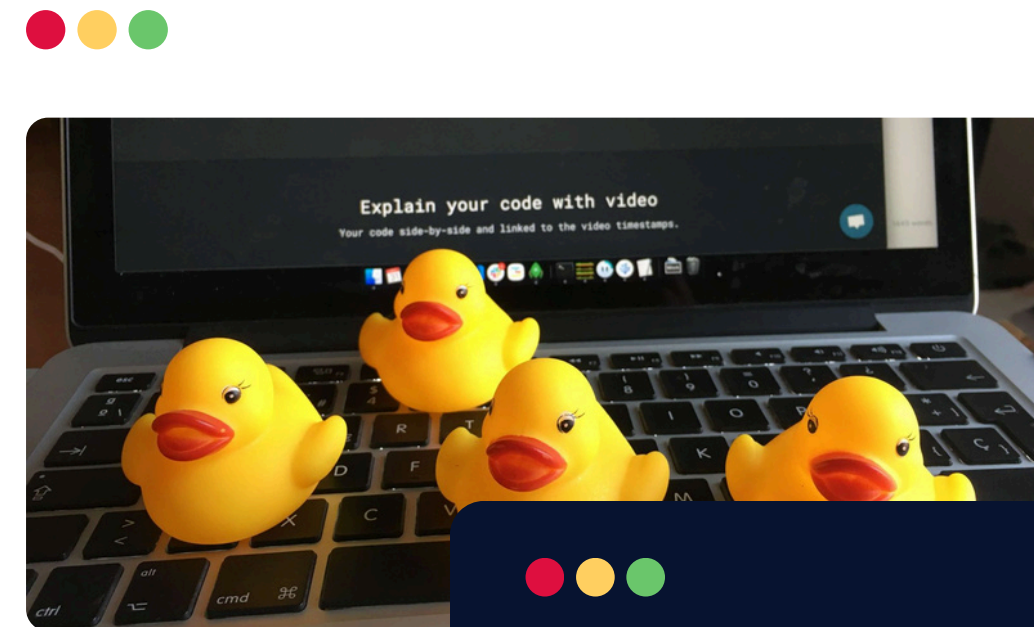
- The polynomial kernel is a type of kernel function that allows the SVM to model non-linear decision boundaries by adding polynomial interaction terms between the input features.

$$K(x, x') = (x^T x' + c)^d$$

- x and x' are the inputs.
- d is the degree of polynomial

Polynomial Kernel





Thank You

created by:
Ahd Ehab